RAG Retrieval Benchmark: A Comprehensive Evaluation of Embedding Models

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Abstract

This paper presents a systematic evaluation framework for embedding models in Retrieval Augmented Generation (RAG) systems. We developed an automated pipeline that processes PDF documents, generates synthetic queries using large language models, and evaluates retrieval performance across multiple embedding models using standardized information retrieval metrics. Our framework provides researchers and practitioners with a reproducible methodology for selecting optimal embedding models for their specific document collections and use cases.

Executive Summary

This report presents a comparative analysis of embedding models for retrieval augmented generation (RAG). The models were evaluated on their ability to retrieve relevant documents using ChromaDB as the vector database. The benchmark used **232 documents** and **10 queries** with **49 relevant document mappings**.

Best performing model: `sentence-transformers/distiluse-base-multilingual-cased-v1` * MAP@3: **0.6111** * Recall@3: **0.3445** * Precision@3: **0.6667** * NDCG@3: **0.6531** * MRR@3: **0.7000**

Methodology

1. Automated Pipeline Overview

Our evaluation framework consists of five main stages:

1. **PDF Processing**: Document ingestion and chunking
2. **Corpus Generation**: Creation of standardized document chunks
3. **Query Generation**: LLM-assisted query synthesis
4. **Relevance Assessment**: Automated qrels generation
5. **Embedding Evaluation**: Multi-model performance comparison

2. Document Processing Pipeline

2.1 PDF Extraction and Chunking

- **Source documents**: PDF files from `utils/pdfs` directory
- **Chunking strategy**: Recursive character text splitting
- **Chunk size**: 1000 characters
- **Chunk overlap**: 200 characters
- **Total chunks generated**: 232

The chunking process uses LangChain's `RecursiveCharacterTextSplitter` to maintain semantic coherence while ensuring manageable chunk sizes for embedding models.

2.2 Query Generation Process

- **LLM Model**: gemini-1.5-flash
- **Number of queries generated**: 10
- **Generation strategy**: Seed-chunk → Query → Retrieval → Label
- **Candidate pool size**: 10 chunks per query

Queries were generated using the following systematic approach: 1. Random sampling of seed chunks from the corpus 2. LLM-based question generation targeting specific chunks 3. Retrieval-based candidate identification using ChromaDB 4. LLM-based relevance labeling for comprehensive qrels

3. Dataset Statistics

3.1 Corpus Characteristics

Metric	Value
Total Documents	232
Average Chunk Length (words)	124.5
Median Chunk Length (words)	121
Min Chunk Length (words)	91
Max Chunk Length (words)	179
Standard Deviation	17

3.2 Query Characteristics

Metric	Value
Total Queries	10
Average Query Length (words)	19.7
Median Query Length (words)	20.5
Min Query Length (words)	12
Max Query Length (words)	27
Standard Deviation	5.2

3.3 Relevance Distribution

Metric	Value
Total Relevance Judgments	49
Average Relevant Docs per Query	4.9

Metric	Value
Median Relevant Docs per Query	5.0
Min Relevant Docs	1
Max Relevant Docs	9
Coverage Ratio	21.12%

4. Experimental Setup

4.1 Evaluation Framework

- **Vector Database**: ChromaDB
- **Evaluation Library**: SentenceTransformers InformationRetrievalEvaluator
- **Retrieval Depth (k)**: 3
- **Batch Size**: 32
- **Collection Reinitialization**: True

4.2 Evaluated Models

Model Name	Collection	Dimensions	Type
sentence-transformers/distiluse-base-multilingual-cased-v1	coll_v1	512	Sentence Transformer
sentence-transformers/all-MiniLM-L6-v2	coll_v3	384	Sentence Transformer
sentence-transformers/distiluse-base-multilingual-cased-v2	coll_v2	512	Sentence Transformer

4.3 Evaluation Metrics

Metric	Description	Measures
MAP@k	Mean Average Precision at rank k	Ranking quality
Recall@k	Proportion of relevant docs in top k	Coverage
Precision@k	Proportion of relevant docs among retrieved	Accuracy
NDCG@k	Normalized Discounted Cumulative Gain	Ranking with position bias
MRR@k	Mean Reciprocal Rank	First relevant position

4.4 Representative Query Examples

Query ID	Query Text	# Relevant Docs
q1_seedChunk_Sample_Pdf_chunk39	İş Planı 2 (5639) için bulut, DevOps, mikroservis mimarisi, ölçeklenebilirlik, veri işleme optimi...	5
q2_seedChunk_Sample_Pdf_chunk62	Here's a search query in Turkish, reflecting the main topics of the provided document: "Yapay zeka ...	7
q3_seedChunk_eajw-54-3-248_chunk38	What are the most common pre-trained models (e.g., U-Net, DeepLabv3) used in a recent competition, a...	4
q4_seedChunk_Sample_Pdf_chunk80	Yapay zeka tabanlı teslim evrakı doğrulama, operasyon optimizasyonu ve raporlama sistemlerinin geliş...	9
q5_seedChunk_Sample_Pdf_chunk84	Veri mühendisliği optimizasyonu, büyük veri işleme altyapısı ve güvenli bulut depolama çözümleri içeren bi...	6

Results

5.1 Model Performance Comparison

Table: Comprehensive Model Performance Results

Model	MAP@3	Recall@3	Precision@3	NDCG@3	MRR@3
sentence-transformers/distiluse-base-multilingual-cased-v1	0.6111	0.3445	0.6667	0.6531	0.7
sentence-transformers/distiluse-base-multilingual-cased-v2	0.6111	0.3269	0.6333	0.6469	0.7333
sentence-transformers/all-MiniLM-L6-v2	0.15	0.1237	0.2333	0.2408	0.45

5.2 Performance Analysis

The evaluation reveals significant performance variations across embedding models. **sentence-transformers/distiluse-base-multilingual-cased-v1** achieved the highest MAP@3 score of 0.6111, while **sentence-transformers/all-MiniLM-L6-v2** scored 0.1500, representing a **307.4% improvement**.

MAP-Recall Correlation: 0.997 (indicates how well ranking quality correlates with coverage)

Model Performance Overview

Figure 1: Model Performance Overview

Corpus Statistics

Figure 2: Corpus Statistics

5.3 Visual Analysis

5.3.1 Overall Performance Comparison *Figure: Comprehensive comparison of all embedding models across evaluation metrics.*

5.3.2 Dataset Characteristics *Figure: Distribution of chunk lengths, query lengths, and relevance patterns in the evaluation dataset.*

5.3.3 Individual Metric Analysis

MAP@3 Results

RECALL@3 Results

PRECISION@3 Results

NDCG@3 Results

MRR@3 Results

5.4 Statistical Significance Analysis

The following tables show absolute differences between model pairs. Differences greater than 0.05 may indicate practically significant improvements for real-world applications.

MAP@3 Pairwise Differences

Map Comparison

Figure 3: Map Comparison

Recall Comparison

Figure 4: Recall Comparison

Precision Comparison

Figure 5: Precision Comparison

Model	distiluse-base-multilingual-cased-v1	distiluse-base-multilingual-cased-v2	all-MiniLM-L6-v2
distiluse-base-multilingual-cased-v1	—	0.0000	0.4611*
distiluse-base-multilingual-cased-v2	0.0000	—	0.4611*
all-MiniLM-L6-v2	0.4611*	0.4611*	—

Note: Values marked with indicate potentially significant differences (>0.05).*

Recall@3 Pairwise Differences

Model	distiluse-base-multilingual-cased-v1	distiluse-base-multilingual-cased-v2	all-MiniLM-L6-v2
distiluse-base-multilingual-cased-v1	—	0.0176	0.2208*
distiluse-base-multilingual-cased-v2	0.0176	—	0.2032*
all-MiniLM-L6-v2	0.2208*	0.2032*	—

Note: Values marked with indicate potentially significant differences (>0.05).*

Precision@3 Pairwise Differences

Ndcg Comparison

Figure 6: Ndcg Comparison

Mrr Comparison

Figure 7: Mrr Comparison

Model	distiluse-base-multilingual-cased-v1	distiluse-base-multilingual-cased-v2	all-MiniLM-L6-v2
distiluse-base-multilingual-cased-v1	—	0.0333	0.4333*
distiluse-base-multilingual-cased-v2	0.0333	—	0.4000*
all-MiniLM-L6-v2	0.4333*	0.4000*	—

Note: Values marked with indicate potentially significant differences (>0.05).*

NDCG@3 Pairwise Differences

Model	distiluse-base-multilingual-cased-v1	distiluse-base-multilingual-cased-v2	all-MiniLM-L6-v2
distiluse-base-multilingual-cased-v1	—	0.0061	0.4123*
distiluse-base-multilingual-cased-v2	0.0061	—	0.4061*
all-MiniLM-L6-v2	0.4123*	0.4061*	—

Note: Values marked with indicate potentially significant differences (>0.05).*

MRR@3 Pairwise Differences

Model	distiluse-base-multilingual-cased-v1	distiluse-base-multilingual-cased-v2	all-MiniLM-L6-v2
distiluse-base-multilingual-cased-v1	—	0.0333	0.2500*

Model	distiluse-base-multilingual-cased-v1	distiluse-base-multilingual-cased-v2	all-MiniLM-L6-v2
distiluse-base-multilingual-cased-v2	0.0333	—	0.2833*
all-MiniLM-L6-v2	0.2500*	0.2833*	—

Note: Values marked with indicate potentially significant differences (>0.05).*

Technical Implementation

6.1 Pipeline Architecture

The benchmark pipeline implements a modular architecture with the following components:

- **Document Processing Module**: PDF extraction and chunking using PyMuPDF and LangChain
- **LLM Integration**: Gemini API for query generation and relevance assessment
- **Vector Database**: ChromaDB with native embedding functions
- **Evaluation Engine**: SentenceTransformers evaluation framework
- **Reporting System**: Automated report generation with visualizations

6.2 Reproducibility

All experiments can be reproduced using the following commands:

```
# 1. Process PDFs into chunks
python simple_rag_benchmark_pipeline.py --prepare-corpus

# 2. Generate queries and relevance judgments
python simple_rag_benchmark_pipeline.py --generate-qrels

# 3. Run complete benchmark
python simple_rag_benchmark_pipeline.py --all
```

6.3 Embedding Model Specifications

sentence-transformers/distiluse-base-multilingual-cased-v1

- **Collection identifier**: `coll_v1`
- **Embedding dimension**: 512
- **Maximum sequence length**: 128
- **Model architecture**: Transformer-based sentence encoder

sentence-transformers/all-MiniLM-L6-v2

- **Collection identifier**: `coll_v3`
- **Embedding dimension**: 384
- **Maximum sequence length**: 256
- **Model architecture**: Transformer-based sentence encoder

sentence-transformers/distiluse-base-multilingual-cased-v2

- **Collection identifier**: `coll_v2`
- **Embedding dimension**: 512
- **Maximum sequence length**: 128
- **Model architecture**: Transformer-based sentence encoder

Discussion

7.1 Key Findings

1. **Model Performance Hierarchy**: sentence-transformers/distiluse-base-multilingual-cased-v1 emerged as the top performer, achieving superior results across multiple evaluation metrics.
2. **Performance Gaps**: The performance difference between the best and second-best models (0.0000 MAP@3) suggests that model selection significantly impacts retrieval quality.
3. **Automated Evaluation Viability**: The LLM-assisted query and relevance generation process demonstrates the feasibility of automated benchmark creation for domain-specific document collections.

7.2 Limitations and Future Work

- **Single-domain evaluation**: Current results are specific to the evaluated document collection
- **LLM bias**: Query generation and relevance judgments may reflect biases of the generative model
- **Limited query diversity**: Future work should explore multi-turn queries and complex information needs
- **Scalability**: Evaluation should be extended to larger document collections and more diverse domains

Conclusion

This study presents a comprehensive framework for evaluating embedding models in RAG systems. Our automated pipeline successfully generated meaningful evaluation datasets and provided statistically robust comparisons of model performance. The results demonstrate that **sentence-transformers/distiluse-base-multilingual-cased-v1** provides optimal retrieval performance for the

evaluated document collection, with a MAP@3 score of 0.6111. The substantial performance variations across models underscore the importance of systematic evaluation in embedding model selection for production RAG systems.

The open-source nature of our evaluation framework enables researchers and practitioners to adapt the methodology for their specific domains and requirements, contributing to more systematic and reproducible evaluation practices in the RAG community.

This report was automatically generated by the RAG Retrieval Benchmark Pipeline at 2025-06-10_15-26-52. Source code and evaluation data available at the project repository.